# Decision Orchestration: Step-by-Step Annotation of Complex Ontologies that Improves Itself from a Handful of Gold Sentences

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## Abstract

Large language models (LLMs) can help bootstrap annotation when gold data is scarce using guidelines with few examples, but they often struggle with specialized annotation conventions and formats, while prompting alone is not reliable and interpretable on how to systematically revise instructions to correct their errors. We propose *decision orchestration*, a framework that turns annotation guidelines into an executable procedure in which code controls structure and models make local decisions. A reflection module serves as a moderation process, using decision traces and small batches of gold annotations to diagnose errors and iteratively refine the procedure.

On Chinese and English UMR and PARSEME, reflection with just a few annotated sentences improves 12 of 14 procedures using small language models or decision models. On the sampled English PARSEME test set, all reflected procedures outperform the best system from the 2018 shared task.

These findings support the framework's effectiveness and substantial potential for annotation under limited supervision, although gains vary across tasks and languages and do not consistently surpass strong baselines.

## 1 Introduction

Annotating a complex ontology takes two things: an understanding of the structure the scheme describes, and the scheme's own codebook, the conventions that make one annotation right and a plausible alternative wrong (Kim et al., 2026). Early in a project, a new scheme, language or domain has a guideline but little gold. With enough gold, supervised models learn both (Bevilacqua et al., 2021; Bai et al., 2022); with a handful of sentences they learn neither. Large language models bring the first ingredient for free: they understand sentences, events and referents well enough to bootstrap an annotation. What remains is to give them the codebook.

For simple labeling, the guideline fits into the prompt (Gilardi et al., 2023; Tan et al., 2024). Complex schemes pose two challenges: a) *guideline following*: the UMR guidelines exceed 300 pages (Van Gysel et al., 2021), burying relevant rules in long contexts (Liu et al., 2024) and an annotation must satisfy dozens of them at once, where following degrades with each added instruction (Harada et al., 2025). b)*interpretable moderation*: correcting annotation errors requires diagnosing and refining how guidelines are applied (Kim et al., 2026). When one prompt produces the whole annotation, errors are difficult to trace to specific decisions, leaving prompt revisions poorly targeted.

We propose **decision orchestration** (Figure 1), which compiles annotation guidelines into executable steps. Code enforces the output format, annotation inventory and structural constraints; models are consulted only at decision nodes, each with the part of the guideline it needs. Each node returns either a *fragment*, a schema-checked annotation from an LLM, or a *decision*, probabilities over candidate annotations from a decision model (TypeSafe, 2026; OpenAI, 2026; Microsoft, 2026). Each step is explicit and logged, allowing a reflector LLM to trace errors to the very steps that produced them and revise the responsible skills or the harness itself. This enables recursive self-improvement (RSI) of the procedure without changing model weights.

To our knowledge, this is the first work to annotate complex schemes step by step along their guidelines, and we show that it pays twice: (1) errors are diagnosable and can be fixed where they arise, and (2) with very little data, RSI makes the procedure much better than the guideline it started from and than a frontier model given the whole guideline.

We evaluate the framework with small LLMs

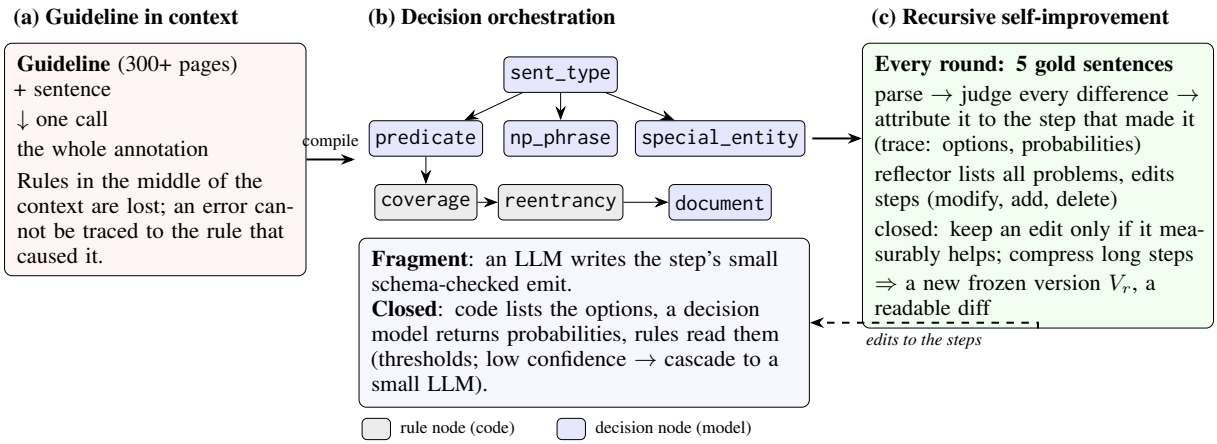



Figure 1: (a) The usual way to give an LLM a codebook: the whole guideline in context. (b) Decision orchestration compiles the guideline into a procedure of small steps, rule nodes and decision nodes, dispatched by a registry; a decision node is either a fragment decision by an LLM or closed questions answered by a decision model. (c) With a handful of gold sentences per round, a reflector edits the steps that made the errors.

and decision models on Chinese and English UMR parsing (Zeman and Štěpánek, 2026) and PARSEME verbal multiword expressions (Ramisch et al., 2018, 2020). Reflection with a few dozen annotated sentences improves most procedures across the two tasks and languages.

Our contributions are threefold:

- **An executable framework for annotation and moderation.** Decision orchestration combines code-enforced constraints with local model judgments. Explicit decision traces support moderation through targeted revisions to skills and the harness, enabling recursive self-improvement without changing model weights.
- **Empirical evidence under limited supervision.** Across Chinese and English UMR and PARSEME, we show that small language models and decision models can perform complex annotation through guideline-derived procedures. These procedures outperform a frontier prompting baseline in some settings, and reflection with 50 annotated sentences improves twelve of fourteen completed procedures.
- **An analysis of how procedures improve and fail.** We trace errors to candidate generation, model judgments and structural rules, and examine the revisions made during reflection. The analysis reveals both concrete opportunities for improvement and limits that local model decisions alone cannot overcome.

Our contributions are:

- **An executable annotation framework.** Decision orchestration turns guidelines into inspectable procedures: code enforces annotation constraints, while small LLMs produce local fragments or decision models assign probabilities to candidate annotations.
- **Moderation through targeted revision.** Step-level records connect errors to the decisions that produced them, enabling a reflector to diagnose failures and revise the responsible skills or the harness itself. This supports recursive self-improvement of the procedure without changing model weights.
- **Evidence of both gains and limits.** Guideline-only procedures outperform a frontier model given the full guideline on Chinese UMR and PARSEME, and reflection improves twelve of fourteen completed procedures by up to 31 F1 points. The gains are task-dependent: the frontier baseline remains stronger on English PARSEME, while closed procedures are limited by the candidates their code can generate. Decision traces expose these limits and identify concrete targets for improvement.

Our contributions are:

- **An executable annotation framework.** Decision orchestration combines code-enforced constraints with annotation fragments from small LLMs or probabilities from decision models. Without annotated examples, its Chinese UMR seeds using decision models outperform a frontier model given the full guideline by 8–11 F1 points.

- **Moderation through traceable decisions.** Step-level records enable a reflector to diagnose errors and revise skills or the harness itself. Our analysis shows how candidate-generation failures limit closed procedures and how measuring proposed edits helps control reflection drift.

- **Improvement with minimal supervision.** Reflection using only 50 annotated sentences improves twelve of fourteen completed procedures, by up to 31 F1 points. On the sampled English PARSEME test set, all reflected procedures outperform the best system from the 2018 shared task, although the frontier prompting baseline remains stronger.

Our findings:

- **Steps beat the guideline in context.** Without any gold, the best seed procedure matches or beats GPT-6.1-sol given the whole guideline on three of four task–language pairs; on Chinese UMR every closed seed is 8–11 points ahead on the official juːmætʃ F1 (Jev 33.9 against 23.9), although a content-based metric, AnCast++, still puts the frontier model ahead (§5.3). On English PARSEME, a scheme close to what frontier models already know, the frontier model stays ahead (§5).

- **A handful of gold goes a long way.** Ten rounds of reflection on five gold sentences each improve twelve of fourteen procedures, by up to 31 points (Jev on English PARSEME, 22.5 → 53.7). The best UMR procedure (Chinese, 37.6) is decided by Jev at under 30 cents per hundred sentences, and on English PARSEME every reflected procedure is above the best shared-task system rescored on the same sentences.

- **Errors have addresses.** The step-level judge shows where the two kinds of decision fail: closed procedures lose where the code offers no right option (e.g., an English lemmatizer that leaves *stolen* unreduced), fragment procedures where structure is left to the model; measured edits keep closed reflection from drifting, unmeasured fragment edits do not (§5.1, §6).

These results suggest that LLMs can learn an annotation scheme from its instructions and a few dozen gold sentences, keep improving as new gold arrives, and that the models doing the annotating need not be frontier models: even simple decision models can serve.

## 2 Related Work

**LLMs as annotators.** LLMs can replace or assist crowd workers on many labeling tasks (Wang et al., 2021; Gilardi et al., 2023; Törnberg, 2023; He et al., 2024; Ziems et al., 2024; Tan et al., 2024), but their quality varies by task and must be validated against human labels (Pangakis et al., 2023). Zero-shot LLM labels often miss a benchmark's specific conventions. Prior work narrows this gap by adding guidelines to prompts and refining them through moderation (Kim et al., 2026), learning guidelines from errors (Pang et al., 2023), or training models to follow guidelines written as code (Sainz et al., 2024; Wang et al., 2023). Yet it remains unclear how a model can apply a long guideline reliably as all of these keep the guideline as one block of context for one model call.

For complex ontologies such as UMR, whose guidelines run to hundreds of pages (Van Gysel et al., 2021; Bonn et al., 2025), that block exceeds what a model reliably uses: performance falls for information in the middle of long contexts (Liu et al., 2024) and drops roughly exponentially with the number of instructions to be followed at once (Harada et al., 2025). Expert-level structured annotation by prompting remains weak (Ettinger et al., 2023; Sun et al., 2024), and supervised parsers need large corpora (Chun and Xue, 2024; Markle et al., 2025). Together, these limitations leave a gap: how to apply long annotation guidelines reliably without a large labeled corpus.

**Recursive self-improvement and rule-based reflection.** Early NLP systems improved annotation by learning rules from gold labels (Brill, 1992, 1995; Ramshaw and Marcus, 1995), but relied on predefined rule templates. LLMs have renewed this idea with more flexible revisions of prompts (Pryzant et al., 2023; Khattab et al., 2024), rule sets (Zhang et al., 2026b; Zhao et al., 2024), or harness codes(Zelikman et al., 2024; Hu et al., 2025; Zhang et al., 2026a). However, their use in expert structured annotation remains underexplored, especially when annotations must follow long, complex guidelines. The limits of self-correction without external feedback (Huang et al., 2024) motivate a structured harness that constrains revisions and checks whether they improve annotation accuracy.

## 3 Methodology: Decision Orchestration

**Workflow and components.** A registry runs rule nodes and decision nodes to produce an annotation and an execution trace. On gold data, a judge scores individual annotation units, and a reflector uses the scores and trace to revise the procedure. We represent this system as

$$\begin{aligned} \Pi &= (\mathcal{S}, R, C, M, W, \Theta, J, \mathcal{F}), \\ s_i &= (I_i, O_i, d_i, T_i) \in \mathcal{S}, \\ \mathcal{F} &= (F_{\text{diag}}, F_{\text{valid}}, F_{\text{comp}}). \end{aligned} \tag{1}$$

Here, $\mathcal{S}$ contains decision nodes $s_i$, each implemented as a skill with input and output schemas $I_i, O_i$, instructions $d_i$, and available tools $T_i$. $R$ is the registry, $C$ the deterministic code, $W$ the worktree, $M$ the annotation and reflection models, and $\Theta$ numerical parameters such as thresholds. $J$ is the set of fine-grained judges; $\mathcal{F}$ comprises diagnosis, edit validation and compression.

$R$ executes nodes in ordered groups (*waves*), dispatching a rule node in $C$ or a decision node in $\mathcal{S}$(*emits*), categorizes outputs to worktree $W$ or to next nodes, and assembles the final outputs from the worktree $W$. At $s_i$, $M$ evaluates the input under $d_i$, with supporting information from $T_i$; $C$ validates or converts its answer to an emit conforming to $O_i$, using $\Theta$ for closed decisions, and assembles the annotation. $J$ scores it against gold in fine-grained aspects, along with a *comprehensive score* defined by the annotation benchmark; $\mathcal{F}$ invokes a reflector in $M$ to locate errors, propose edits to the responsible instructions or code, then validates and compresses the revised procedure to avoid skill instructions from becoming too verbose.

**Modes of Decision nodes.** Decision nodes have two implementations with the same emit. In **fragment mode**, an LLM reads the skill's instructions and input, uses its available tools, and returns a graph fragment in structured outputs. In **closed mode**, the system uses codes to enumerate options and asks choice, yes/no or graded questions. A decision model returns probabilities over these options, which rules convert into the emit using thresholds in $\Theta$. Both modes log inputs and outputs; closed mode also records options and probabilities.

**Cascading** For closed decisions with confidence lower than a *cascading threshold*, a **cascading** mechanism routes the input to a stronger model that receives the same inputs and replaces the original answer.

**Calibration** For the *cascading threshold* above, and to account for different decision models' output probabilities for*decision thresholds*, we fit both in 0.1 increments on a small batch of training annoations for a best comprehensive score.

**Fine-grained evaluation.** The judges in $J$ score individual annotation units against gold; execution traces link predictions to the responsible steps, providing targeted feedback for reflection. Task-specific checks and scoring details are given in the Metrics paragraph of Experiments (§4).

**Reflection.** Reflection revises the procedure in three steps: **Diagnose ($F_{\text{diag}}$):** use judge feedback and execution traces to locate errors and propose targeted edits. **Validate ($F_{\text{valid}}$):** check candidate revisions and retain those that satisfy the acceptance criteria. **Compress ($F_{\text{comp}}$):** merge redundant instructions and rules while preserving required options and conventions. The experimental protocol is detailed in §4 (Reflection protocol).

## 4 Experiments

**Tasks.** We evaluate two structured annotation tasks (Figure 2). **UMR** maps sentences to token-aligned graphs of concepts, roles and attributes, and adds document-level temporal, modal and coreference relations (Van Gysel et al., 2021; Bonn et al., 2025). **PARSEME** identifies verbal multiword expressions (VMWEs), including discontinuous expressions, and assigns each a category (Ramisch et al., 2018, 2020).

**Data.** For UMR, we use Chinese and English data from the DMR 2026 shared task, largely based on UMR 2.1 (Zeman and Štěpánek, 2026). Each language has 50 training sentences in ten batches of five; the test sets contain 100 Chinese and 95 English sentences. We parse test chunks of at most five consecutive sentences independently, using shared-task tokenization. For PARSEME, we use Chinese edition 1.2 (Ramisch et al., 2020) and English edition 1.1 (Ramisch et al., 2018). Each language has 50 training sentences in ten batches of five (three with a VMWE and two without per batch) and 100 sampled official test sentences (50 with a VMWE and 50 without). Table 1 lists the decision skills and rule nodes, Figure 3 shows a UMR trace, and Appendix A gives the seed instructions.

**(a) UMR: a sentence graph, its alignment, and document-level relations**

*And*$_1$ *you*$_2$ *watch*$_3$ *him*$_4$ *pluck*$_5$ *a*$_6$ *few*$_7$ *pears*$_8$ *,*$_9$ (sentence 10 of a narrative)

```
(s10w / watch-01
 :ARG0 (s10p / person
    :refer-person 2nd :refer-number singular)
 :ARG1 (s10p2 / pluck-01
    :ARG0 (s10p3 / person
       :refer-person 3rd :refer-number singular)
    :ARG1 (s10p4 / pear :quant (s10a / a-few))
    :aspect activity)
 :aspect state)
```

```
alignment: s10w 3-3, s10p 2-2, s10p2 5-5,
 s10p3 4-4, s10p4 8-8, s10a 6-7
:temporal ((s8e :after s10w)
       (s10w :overlap s10p2))
:modal ((root :modal author)
     (author :full-affirmative s10w) ...)
:coref ((s8p :same-entity s10p)
     (s9p2 :same-entity s10p4))
```

**(b) PARSEME: verbal multiword expressions**

*After that, he made*$_6$ *a great effort*$_9$ *to be calm.*
$\{6, 9\}$: `LVC.full` (light verb + predicative noun, discontinuous)

*So Harry had been brought*$_5$ *up*$_6$ *by his dead mother's sister ...*
$\{5, 6\}$: `VPC.full` (the particle changes the verb's meaning)

我 已經 作出$_3$ 決定$_4$ ， 讓 他 發言 。
$\{3, 4\}$: `LVC.full`; $\{3\}$: `VPC.semi` (one token: 作 + complement 出)

Categories (EN 1.1): VID, LVC.full, LVC.cause, VPC.full, VPC.semi, IAV;
(ZH 1.2): VID, LVC.full, LVC.cause, VPC.semi, MVC.

Figure 2: Training examples of (a) UMR sentence graphs, token alignments and document-level relations, and (b) PARSEME verbal multiword expressions, which may be discontinuous or overlap; in Chinese, one token can form an expression.

| Scheme | Node | Kind | Responsibility |
|---|---|---|---|
| UMR | `sent_type` | Decision | Classify a span and route it; emit a graph for a leaf. |
| | `discourse` | Decision | Split a span into clauses and assign their discourse relation. |
| | `predicate` | Decision | Choose the head, sense, aspect, modal strength, polarity and argument roles. |
| | `np_phrase` | Decision | Annotate noun-phrase heads, modifiers, roles, quantity and reference attributes. |
| | `special_entity` | Decision | Annotate names, dates, quantities and ordinals. |
| | `document` | Decision | Assign temporal, modal and coreference relations. |
| | `coverage` | Rule | Return uncovered content tokens to sentence routing. |
| | `antecedent` | Rule | Record participants outside the current span for later merging. |
| | `reentrancy` | Rule | Merge nodes with matching types and spans and resolve outside participants. |
| PARSEME | `candidates` | Rule | Propose token groups from forms, lemmas, parts of speech and dependencies. |
| | `decide` | Decision | Identify and categorize expressions: return token groups in fragment mode, or answer linguistic tests in closed mode. |

Table 1: Decision skills use instructions and model $M$; rule nodes are implemented in code $C$. Both schemes use code to assemble the final output.

**Baseline.** GPT-6.1-sol generates the annotation in a single call per test chunk, given the full task guideline, the input sentences (including lemma, part-of-speech and dependency columns for PARSEME) and the required output format, with no gold examples.

**Seeding.** We write the seed procedures by hand from the published guidelines before reading any training gold; neither mode includes worked examples (Appendix A). Both modes use the same task decomposition and rule nodes. In fragment mode, decision skills restate the relevant guideline in natural language and return schema-constrained graph fragments or expressions. In closed mode, we encode the guideline's tests as fixed-choice questions, thresholds and deterministic branches; the model supplies choices and code assembles the annotation. The UMR seeds follow the official file format's inventory of relations and attributes.

**Models and runs.** Fragment decisions use GPT-6-luna and Claude Haiku 5.5 ($0.10 per million input tokens each), with low reasoning effort. Closed decisions use Jev 1.13 (TypeSafe, 2026), OpenAI's Decisions API with `gpt-6-luna-decisions` (OpenAI, 2026) and Microsoft-Decision-1 (Microsoft, 2026), all through the same interface; the cascade re-asks uncertain Jev questions of Haiku 5.5. Claude Sonnet 5.5 reflects in every run. Each configuration runs ten rounds of reflection from its seed, and every frozen version $V_0, \ldots, V_{10}$ is tested. Within our budget and OpenRouter's limit of 20 requests a minute for OpenAI models, this was done for Haiku, Jev and Microsoft on both tasks and for GPT-6-luna on PARSEME; the OpenAI Decisions chains are reported as far as they ran (Limitations). A second, smaller comparison (§5.3) gives every decider two rounds of reflection, including GPT-6-luna and OpenAI Decisions on UMR and a larger fragment decider, Claude Sonnet 5.5 (on UMR through the Claude subscription, on PARSEME through OpenRouter), and tests UMR on 25 test sentences: five chunks (Chinese 0, 4, 8, 12, 16; English 0, 3, 7, 10, 18, leaving out `en-0007`,

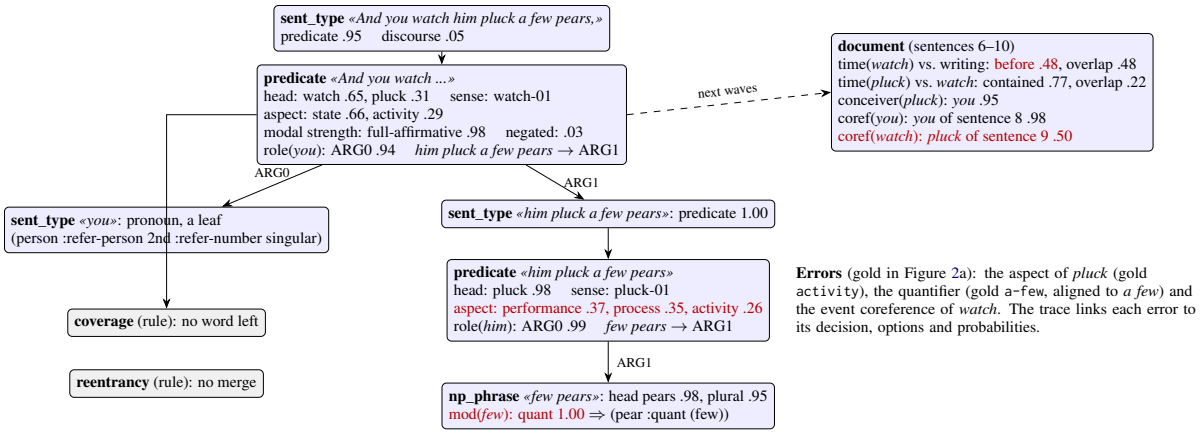



Figure 3: A real trace of the closed-mode UMR seed with Jev. Blue: decision nodes with options and probabilities; grey: rule nodes; red: decisions that differ from gold (Figure 2a).

which repeats the text of `0005`). Every other system is re-scored on the same chunks from its per-chunk counts, and PARSEME uses the full test sets.

**Reflection protocol.** Each round parses one batch of five gold sentences. The reflector reads gold and predicted annotations, unit-level scores, attributed errors, decision traces and editable files, then proposes additions, replacements or deletions (Algorithm 1). In closed mode, each edit is applied to a copy and evaluated on the current and all earlier training batches. It is kept if current-batch F improves and earlier-batch F falls by at most 0.5, or earlier-batch F improves without a current-batch loss. Rejected edits are returned once with their measured effects. In fragment mode, edits are accepted if they apply and leave the procedure loadable, then tested in the next round.

Files exceeding $1.5\times$ their seed length are compressed, merging redundant rules while preserving options, questions and conventions. In closed mode, a compressed version is kept only if neither batch set loses more than 0.5 F. Each round freezes the revised procedure with a reviewable diff. Edits copying six consecutive Chinese characters or four English words from test text are rejected.

**Metrics.** UMR is scored with the shared task's official juːmætʃ F1 (Zeman and Štěpánek, 2026): predicted nodes are mapped to gold nodes through their token alignment, and F1 is computed over the triples (concepts, relations, attributes) of mapped nodes, document-level relations included; we also report it without the document level ("no doc"). For diagnosis we use a fine-grained judge (Table 2): it maps nodes with AnCast (Sun and Xue, 2024), breaks the comparison into 25 checks in 11 modules, gives every unit a verdict and every check its own F1, and attributes each predicted unit to the skill call that produced it. PARSEME is scored with the official evaluation script: MWE-based F1 (exact token set and category), token-based F1, and F1 on expressions unseen in the training data. A PARSEME judge sorts every gold and predicted expression into six kinds: correct, wrong category (same tokens), partial (overlapping tokens), missed without a candidate (no rule proposed its tokens), missed after rejection (a candidate had them), and spurious (no overlap with any gold expression).

**State of the art.** For UMR we compare with the best shared-task submission (Zeman and Štěpánek, 2026), entered as a placeholder of 50 F in both languages until it is re-scored on our test chunks. For PARSEME, the best systems of the shared tasks (Chinese 1.2: TRAVIS-mono/-multi, open track; English 1.1: Milos, open track, and TRAPACC, closed track), re-scored on our sentences, and the best published English 1.1 result (Rohanian et al., 2019), quoted on the full test set.

## 5 Results

Table 3 gives the headline numbers and Figure 4 the test score of every version along the way. Appendix Table 9 lists every number.

**The guideline as steps beats the guideline in context.** With no gold data at all, the seed procedures already compete with a frontier model that reads the whole guideline. On Chinese UMR, every closed seed (Jev 33.9, Microsoft 35.0, OpenAI 32.1) is 8–11 points above GPT-6.1-sol (23.9), although each decision is taken by a model that cannot write text; on English UMR the seeds are

| Module | Checks | What one unit compares (for matched gold/predicted nodes) | Unit kind |
|---|---|---|---|
| a. nodes | event found, entity found | does a gold event (entity) node have a predicted match; is a predicted one matched | node |
| | event concept, entity concept | is the concept the same | node pair |
| | granularity | is the node aligned to the same words (one node for several gold nodes or vice versa) | span |
| b. sense | event sense | same PropBank roleset number, once the concept is right | node pair |
| c. attributes | aspect, modal strength, polarity, refer, other | same value of the attribute; an attribute only one side has is missing or extra | attribute |
| d. arguments | core args of events / of abstract predicates; peripheral args of events / of entities | how far apart the argument sets of a matched head are (credit = share that agrees) | head |
| e. role labels | core, peripheral | same role label on a matched edge | edge |
| f. names | name tokens, wiki | same `:op` tokens of a matched name; same wiki title | entity |
| g. reentrancy | reentrancy | is an edge back to an already declared node also in gold | edge |
| h. temporal | temporal dependency | a child node's temporal parent(s) and relation(s) | child |
| i. modal | modal dependency | an event's conceiver (parent) and modal strength (label) | child |
| j. coreference | chains, subsets | is a mention linked to the same other mentions as in gold; subset relations | mention |
| k. format | document format | are the predicted document-level relations well formed (no gold needed) | relation |

Table 2: The fine-grained UMR judge. Every unit gets a verdict: correct (credit 1), partial (0.5, or the check's own credit), wrong, missing or extra (0). Recall counts the gold side (correct, partial, wrong, missing), precision the predicted side (correct, partial, wrong, extra), and each check reports its F1. Modules a–f never see reentrant edges, so module g's errors do not move them. Each predicted unit is attributed to the skill call that wrote it.

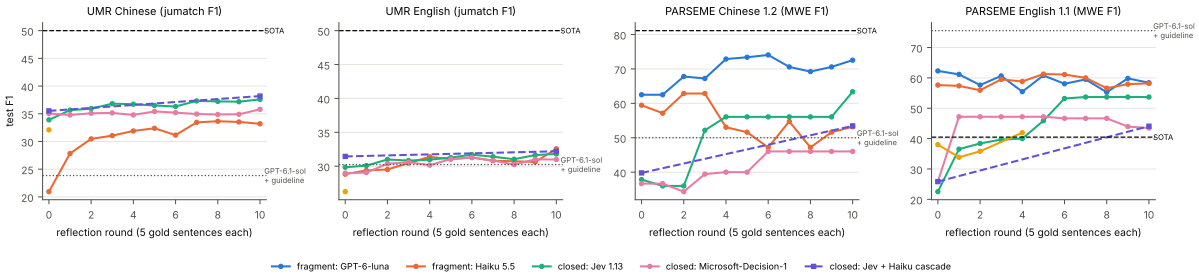



Figure 4: Test F1 of every frozen version $V_0, \ldots, V_{10}$; round $r$ has read $5r$ gold sentences. Dotted: GPT-6.1-sol with the whole guideline; dashed: best shared-task system (UMR: placeholder).

within 1.5 points of it (29.0–29.8 against 30.2), the small LLMs as fragment deciders on par (Haiku 28.8). The one UMR seed far below the baseline, Haiku on Chinese (20.9), fails in a way that the trace shows at once: it translated every Chinese concept into English (计划 → `plan-01`), which no guideline sentence forbids explicitly (event concepts 30.0 F in the judge, Table 4). On PARSEME, the fragment seeds start well above the frontier baseline in Chinese (GPT-6-luna 62.5, Haiku 59.5, against 50.0), while the closed seeds start low (22.5–37.8), for a reason we discuss below; in English every seed is below the frontier baseline (at most 62.3 against 75.6).

**Fifty gold sentences of reflection.** Reflection improves twelve of the fourteen chains that finished. On Chinese UMR, Haiku climbs from 20.9 to 33.2 (best 33.6), the first round alone adding 6.9 points by writing the missing convention ("for Chinese the concept is the source word itself, with a sense number"), and Jev from 33.9 to 37.6, the best UMR procedure without a cascade, 13.7 points above the frontier baseline. On English UMR the gains are smaller: Haiku 28.8 → 32.6, Jev 29.8 → 31.8, Microsoft 29.0 → 31.0, all above the baseline at $V_{10}$. On PARSEME, the measured closed chains gain most: Jev rises from 37.8 to 63.4 in Chinese and from 22.5 to 53.7 in English, Microsoft from 36.7 to 46.0 and from 25.9 to 43.5, and OpenAI Decisions, in the four English rounds it completed, from 38.0 to 41.9. The fragment chains on PARSEME, whose edits are not measured, move less predictably: GPT-6-luna rises in Chinese from 62.5 to 72.5 (best 74.1 at $V_6$), the best Chinese PARSEME result and 22.5 points above the frontier baseline, but in English it stays between 55.3 and 62.3 and ends at 58.4; Haiku reaches 62.9 in Chinese at $V_3$ and drifts back to 53.2 by $V_{10}$ (§6).

**Against the state of the art.** On English PARSEME, every procedure after reflection is above the best system of the 1.1 shared task rescored on the same sentences (Milos, 40.5): Jev at 53.7, Haiku at 58.2, GPT-6-luna at 58.4, from 50 annotated sentences against the full training data. The English 1.1 systems predate pretrained language models, however, and the frontier model given the guideline is better still (75.6): in English, which these models know best, the guideline in context works, and decomposition with small deciders does not catch up within ten rounds. On Chinese

| System | UMR (juːmætʃ F1) Chinese | UMR (juːmætʃ F1) English | PARSEME (MWE F1) Chinese | PARSEME (MWE F1) English |
|---|---|---|---|---|
| Best shared-task system | 50.0† | 50.0† | 81.1 | 40.5 |
| GPT-6.1-sol + full guideline | 23.9 | 30.2 | 50.0 | 75.6 |
| *fragment*: GPT-6-luna, seed $V_0$ | – | – | 62.5 | 62.3 |
| $V_{10}$ | – | – | 72.5 | 58.4 |
| best version | – | – | 74.1 ($V_6$) | 62.3 ($V_0$) |
| *fragment*: Haiku 5.5, seed $V_0$ | 20.9 | 28.8 | 59.5 | 57.6 |
| $V_{10}$ | 33.2 | 32.6 | 53.2 | 58.2 |
| best version | 33.6 ($V_8$) | 32.6 ($V_{10}$) | 62.9 ($V_3$) | 61.3 ($V_5$) |
| *closed*: Jev 1.13, seed $V_0$ | 33.9 | 29.8 | 37.8 | 22.5 |
| $V_{10}$ | 37.6 | 31.8 | 63.4 | 53.7 |
| best version | 37.6 ($V_{10}$) | 31.8 ($V_{10}$) | 63.4 ($V_{10}$) | 53.7 ($V_{10}$) |
| *closed*: OpenAI Decisions, seed $V_0$ | 32.1 | 26.2 | – | 38.0 |
| $V_{10}$ | – | – | – | 41.9 ($V_4$) |
| best version | 32.1 ($V_0$) | 26.2 ($V_0$) | – | 41.9 ($V_4$) |
| *closed*: Microsoft-Decision-1, seed $V_0$ | 35.0 | 29.0 | 36.7 | 25.9 |
| $V_{10}$ | 35.8 | 31.0 | 46.0 | 43.5 |
| best version | 35.8 ($V_{10}$) | 31.3 ($V_6$) | 46.0 ($V_{10}$) | 47.2 ($V_5$) |
| Jev $V_{10}$, calibrated | 37.4 | 31.8 | 61.4 | 49.5 |
| Jev $V_{10}$ + Haiku cascade | 38.1 | 32.6 | 53.4 | 44.1 |

Table 3: Test F1. UMR: the shared task's juːmætʃ metric on 100 Chinese and 95 English sentences; PARSEME: MWE-based F1 on 100 sentences per language. $V_0$ is the seed written from the guideline, $V_{10}$ the procedure after ten rounds of reflection on five gold sentences each (50 sentences in all); "best version" is the best of $V_0..V_{10}$ on the test set and is shown only to describe the curves (Figure 4). †Placeholder until the best shared-task submission is re-scored on our chunks. PARSEME shared-task systems are trained on the full training data and re-scored on our sentences: TRAVIS-multi (Chinese 1.2) and Milos (English 1.1). Cascade: uncertain Jev answers re-asked of Haiku 5.5, $\tau$ as in Table 6. Red dashes: runs not finished within our budget (see Limitations; GPT-6-luna and OpenAI Decisions on UMR after two rounds are in Table 7); a version in parentheses is the last one a stopped chain reached.

| Check | Chinese Haiku (fragm.) $V_0$ | Chinese Haiku (fragm.) $V_{10}$ | Chinese Jev (closed) $V_0$ | Chinese Jev (closed) $V_{10}$ | English Haiku (fragm.) $V_0$ | English Haiku (fragm.) $V_{10}$ | English Jev (closed) $V_0$ | English Jev (closed) $V_{10}$ |
|---|---|---|---|---|---|---|---|---|
| event concept | 30.0 | 77.3 | 73.5 | 77.9 | 66.7 | 72.5 | 49.4 | 55.6 |
| entity concept | 36.3 | 73.8 | 64.4 | 69.7 | 69.5 | 76.3 | 68.1 | 72.5 |
| granularity | 82.5 | 85.8 | 92.2 | 92.9 | 84.9 | 88.1 | 90.3 | 90.7 |
| event sense | 58.6 | 84.7 | 84.4 | 86.1 | 63.2 | 71.9 | 56.8 | 60.9 |
| aspect | 60.5 | 65.7 | 51.7 | 63.1 | 52.0 | 61.3 | 55.2 | 57.5 |
| polarity | 52.6 | 61.1 | 56.4 | 49.0 | 36.8 | 36.8 | 32.4 | 37.8 |
| refer (person/number) | 20.5 | 23.2 | 42.6 | 49.0 | 35.3 | 49.9 | 35.8 | 37.7 |
| core args of events | 22.2 | 45.6 | 37.8 | 35.7 | 27.5 | 29.5 | 23.0 | 24.9 |
| peripheral args of events | 12.3 | 25.3 | 15.4 | 20.9 | 9.0 | 17.6 | 16.8 | 14.0 |
| core role labels | 60.2 | 75.1 | 74.8 | 68.6 | 65.4 | 60.9 | 69.7 | 71.8 |
| name tokens | 50.3 | 52.0 | 46.3 | 52.2 | 61.6 | 67.6 | 45.4 | 46.9 |
| reentrancy | 9.7 | 24.4 | 18.3 | 20.4 | 11.4 | 12.6 | 9.8 | 10.5 |
| temporal | 4.5 | 11.1 | 16.1 | 18.5 | 6.7 | 14.5 | 13.7 | 16.8 |
| modal | 48.7 | 54.6 | 55.2 | 56.5 | 28.6 | 31.7 | 28.1 | 27.2 |
| coreference | 3.0 | 16.4 | 4.3 | 12.7 | 11.9 | 22.5 | 9.3 | 11.3 |
| juːmætʃ F1 (official) | 20.9 | 33.2 | 33.9 | 37.6 | 28.8 | 32.6 | 29.8 | 31.8 |

Table 4: The fine-grained UMR judge (F1 per check, %) for fragment decisions by Haiku and closed decisions by Jev, seed and $V_{10}$. Selected checks; all 25 are in the released results.

PARSEME the 1.2 systems, trained on the full data with pretrained encoders, remain ahead (TRAVIS-multi 81.1 against GPT-6-luna's 72.5), although our best procedure is on par with the third open-track system (MTLB-STRUCT, 73.1) and above the best closed-track one (ERMI, 62.8). On UMR, our best procedures are 12–18 points below the placeholder for the best shared-task system.

### 5.1 Where fragment and closed decisions differ

Because every unit of an annotation is attributed to the step that made it, the two modes can be compared check by check (Table 4). On UMR the closed procedure wins on what code controls: *granularity* (which words a node covers; Chinese 92.9 against 85.8), *refer* (person and number of pronouns, filled from closed-class tables; 49.0 against 23.2) and temporal relations. The fragment procedure wins on what needs open-ended judgment: the arguments of events (Chinese core arguments 45.6 against 35.7), reentrancies (24.4 against 20.4) and coreference chains (16.4 against 12.7). juːmætʃ maps nodes through their token alignment, so granularity errors cost twice, and Jev comes out ahead in Chinese (37.6 against 33.2).

In English the order flips (Haiku 32.6, Jev 31.8), and the judge says why: event concepts (72.5 against 55.6) and senses (71.9 against 60.9). The trace shows that much of the difference is not in a decision at all. Jev gets 212 English event concepts wrong against Haiku's 136, and 51 of Jev's are inflected forms (*stolen*, *thought*, *fallen*, *trie*) that the seed's rule-based lemmatizer failed to reduce, against 20 for Haiku, which lemmatizes as it writes; in Chinese, without inflection, both have 107 and almost none is of this kind. A decision model chooses among the options that code gives it, so the closed mode is exactly as good as its enumerator: when the right concept is not an option, no probability can rescue it. Such an error is cheap to fix once found (an irregular-verb table), and it is found by reading one column of the judge's output.

On PARSEME the PARSEME judge (Table 5) explains why the closed seeds start so low: they predict 121–175 expressions for 57–59 gold, of which 78–123 are spurious. Every closed question is a test from the guideline's decision trees, answered by a model that has never seen the scheme's sense of "idiomatic" or "light"; at the seed's threshold of 0.5, far too many candidates pass. Reflection is where the closed procedure learns that sense: Jev's spurious predictions fall from 80 to 9 in Chinese and from 112 to 19 in English, through edits such as "有 is not a light verb", "no VID for single-character object nouns" or "drop IAV on a verb that

| Lang. | System | correct | wrong cat. | partial | missed, no cand. | missed, rejected | spurious | F |
|---|---|---|---|---|---|---|---|---|
| Chinese | GPT-6.1-sol | 25 | 3 | 4 | 27 | 0 | 21 | 50.0 |
| | GPT-6-luna $V_0$ | 23 | 12 | 1 | 6 | 17 | 17 | 62.5 |
| | GPT-6-luna $V_{10}$ | 33 | 4 | 1 | 6 | 15 | 5 | 72.5 |
| | Haiku 5.5 $V_0$ | 26 | 7 | 4 | 3 | 19 | 16 | 59.5 |
| | Haiku 5.5 $V_{10}$ | 22 | 3 | 3 | 5 | 26 | 8 | 53.2 |
| | Jev 1.13 $V_0$ | 7 | 28 | 12 | 8 | 4 | 80 | 37.8 |
| | Jev 1.13 $V_{10}$ | 22 | 10 | 1 | 11 | 15 | 9 | 63.4 |
| | Microsoft-Decision-1 $V_0$ | 9 | 24 | 11 | 8 | 7 | 78 | 36.7 |
| | Microsoft-Decision-1 $V_{10}$ | 16 | 13 | 5 | 8 | 17 | 34 | 46.0 |
| English | GPT-6.1-sol | 46 | 5 | 4 | 2 | 0 | 22 | 75.6 |
| | GPT-6-luna $V_0$ | 31 | 7 | 8 | 6 | 5 | 21 | 62.3 |
| | GPT-6-luna $V_{10}$ | 27 | 6 | 8 | 7 | 9 | 17 | 58.4 |
| | Haiku 5.5 $V_0$ | 30 | 4 | 8 | 7 | 8 | 19 | 57.6 |
| | Haiku 5.5 $V_{10}$ | 25 | 7 | 9 | 6 | 10 | 14 | 58.2 |
| | Jev 1.13 $V_0$ | 19 | 6 | 20 | 10 | 2 | 112 | 22.5 |
| | Jev 1.13 $V_{10}$ | 23 | 6 | 4 | 16 | 8 | 19 | 53.7 |
| | OpenAI Decisions $V_0$ | 11 | 12 | 9 | 18 | 7 | 29 | 38.0 |
| | Microsoft-Decision-1 $V_0$ | 17 | 13 | 15 | 11 | 1 | 123 | 25.9 |
| | Microsoft-Decision-1 $V_{10}$ | 10 | 10 | 5 | 18 | 14 | 11 | 43.5 |

Table 5: PARSEME error kinds on the test set (counts of gold or predicted expressions): correct; same tokens with the wrong category; overlapping tokens; gold missed because no rule proposed its tokens, or because a proposed candidate was rejected; spurious predictions. F is MWE-based, category-blind as in the official script.

already has a VPC", each measured on the training batches before it was kept. What the closed procedure cannot fix this way are expressions no rule proposes ("missed, no candidate": 11 in Chinese, 16 in English, against 3–7 for the fragment procedures, which may return expressions outside the candidate list).

### 5.2 Calibration, cascade and transfer

Table 6 shows the two closed-mode adjustments of §3. **Threshold calibration** helps six of the eight seeds, a little on UMR (Jev 33.9 → 34.2 and 29.8 → 30.2, Microsoft Chinese 35.0 → 35.8) and more on PARSEME (Jev English 22.5 → 27.1, Chinese 37.8 → 39.2, Microsoft English 25.9 → 29.9), but not the reflected procedures (UMR within $\pm 0.2$; PARSEME Jev 63.4 → 61.4 and 53.7 → 49.5): after ten rounds of measured reflection, the thresholds have already been tuned implicitly by the edits that were kept (a "lower idiom threshold" edit, for example), and the remaining training gain of calibration (Jev Chinese 88.2 → 89.6) does not carry over to the test set (63.4 → 61.4). With 50 sentences, thresholds fitted on the training data overfit quickly. **The cascade** routes questions Jev answers with confidence below $\tau$ to Haiku. On PARSEME it hurts: on the training sentences no $\tau$ improved Jev's $V_{10}$ (English 76.7 without the cascade, 64.6–73.5 with it; Chinese 88.2 against 77.3–82.2), and at $\tau = 0.7$ the test score falls too (English 53.7 → 44.1, Chinese 63.4 → 53.4), while on the seeds it helps a little (37.8 → 39.8, 22.5 → 25.8). Once reflection has tuned PARSEME's rules to Jev's answers, replacing the hard ones with another model's breaks that fit. On UMR, whose closed questions are more open (which concept, which role), the training sentences select $\tau = 0.8$ in both languages (Chinese 37.5 → 37.8, English 38.5 → 38.8), and the cascade adds a little on the test set too: Chinese 37.6 → 38.1, the best UMR result, and English 31.8 → 32.6, level with Haiku's fragment procedure, for $0.18–0.67 per hundred sentences, still a fraction of Haiku's own cost. On the seeds it adds 1.6 points in both languages (33.9 → 35.5, 29.8 → 31.4). **Transfer.** The procedure Jev's chain learned works with another decision model, and in three of four cases better than the procedure that model's own chain learned. Answered by Microsoft-Decision-1, Jev's $V_{10}$ scores 57.1 on Chinese PARSEME against 46.0 for Microsoft's own $V_{10}$, and 62.0, close to Jev's 63.4, once its thresholds are calibrated for Microsoft, which is the situation calibration is meant for; 36.6 against 35.8 on Chinese UMR; 44.2 against 43.5 on English PARSEME; only on English UMR is it lower (30.4 against 31.0). What reflection writes is mostly about the scheme, not about the model, and Jev's chain learned more of it than Microsoft's.

### 5.3 Two rounds with every decider

Table 7 gives every decider the same small budget, two rounds of reflection, and fills the cells the full runs could not reach. **A stronger fragment decider** helps most. Claude Sonnet 5.5 has the best fragment seeds and, after two rounds, the best UMR procedure in English (42.5 juːmætʃ, against 37.1 for GPT-6.1-sol and 38.0 for Haiku at $V_2$) and the best Chinese PARSEME procedure of the paper (78.3, three points below TRAVIS-multi's 81.1, from ten annotated sentences against the full training data); in Chinese UMR it is level with Jev on juːmætʃ (35.0 against 35.7). English PARSEME stays where every fragment decider leaves it (56–59). **GPT-6-luna** starts lowest on UMR (Chinese 18.9, English 24.4): like Haiku's seed, it writes English concepts for Chinese words (event concepts 15.7 F in the judge), and two rounds add about ten points in both languages (Chinese 18.9 → 28.4, English 24.4 → 34.3), the largest two-round gains in the table. **OpenAI Decisions** gains little in two rounds (English UMR 30.4 → 31.2; the Chinese chains did not finish under the rate limit; English PARSEME 38.0 → 35.8, and 41.9 after four rounds).

**The metric matters.** AnCast++ matches nodes by their content and neighbourhood, juːmætʃ through their token alignment. On the same predictions the two agree that ten rounds of reflec-

| Procedure | UMR zh raw | UMR zh cal. | UMR en raw | UMR en cal. | PARSEME zh raw | PARSEME zh cal. | PARSEME en raw | PARSEME en cal. |
|---|---|---|---|---|---|---|---|---|
| Jev 1.13 $V_0$ | 33.9 | 34.2 | 29.8 | 30.2 | 37.8 | 39.2 | 22.5 | 27.1 |
| Jev 1.13 $V_{10}$ | 37.6 | 37.4 | 31.8 | 31.8 | 63.4 | 61.4 | 53.7 | 49.5 |
| Microsoft-Decision-1 $V_0$ | 35.0 | 35.8 | 29.0 | 29.0 | 36.7 | 35.0 | 25.9 | 29.9 |
| Microsoft-Decision-1 $V_{10}$ | 35.8 | 36.0 | 31.0 | 31.1 | 46.0 | 46.0 | 43.5 | 43.5 |
| Jev $V_{10}$ answered by Microsoft | 36.6 | – | 30.4 | – | 57.1 | 62.0 | 44.2 | 42.3 |
| Jev $V_0$ / + cascade | 33.9 | 35.5 | 29.8 | 31.4 | 37.8 | 39.8 | 22.5 | 25.8 |
| Jev $V_{10}$ / + cascade | 37.6 | 38.1 | 31.8 | 32.6 | 63.4 | 53.4 | 53.7 | 44.1 |

Table 6: Closed procedures with their thresholds as written (raw) and calibrated on the 50 training sentences (cal.); Jev's $V_{10}$ answered by Microsoft-Decision-1 (on UMR not calibrated, for budget); and Jev with the cascade to Haiku 5.5, at the $\tau$ selected on the training sentences (UMR: 0.8) or, where no $\tau$ helped there (PARSEME), at $\tau = 0.7$.

| Decisions | | UMR zh (25 sent.) juːmætʃ | UMR zh (25 sent.) AnCast | UMR en (25 sent.) juːmætʃ | UMR en (25 sent.) AnCast | PARSEME zh | PARSEME en |
|---|---|---|---|---|---|---|---|
| GPT-6.1-sol + guideline | | 22.6 | 43.3 | 37.1 | 55.9 | 50.0 | 75.6 |
| *fragment*: Haiku 5.5 | $V_0$ | 22.2 | 22.3 | 35.8 | 38.3 | 59.5 | 57.6 |
| | $V_2$ | 32.3 | 39.2 | 38.0 | 38.2 | 62.9 | 55.9 |
| *fragment*: GPT-6-luna | $V_0$ | 18.9 | 17.2 | 24.3 | 22.4 | 62.5 | 62.3 |
| | $V_2$ | 28.4 | 35.8 | 34.3 | 35.4 | 67.8 | 57.6 |
| *fragment*: Sonnet 5.5 | $V_0$ | 29.7 | 36.6 | 39.7 | 43.4 | 72.1 | 58.5 |
| | $V_2$ | 35.0 | 41.1 | 42.5 | 51.6 | 78.3 | 56.9 |
| *closed*: Jev 1.13 | $V_0$ | 33.3 | 30.3 | 34.5 | 30.1 | 37.8 | 22.5 |
| | $V_2$ | 35.7 | 31.9 | 35.6 | 29.8 | 36.0 | 38.4 |
| *closed*: OpenAI Decisions | $V_0$ | 33.9 | 27.1 | 30.4 | 24.2 | – | 38.0 |
| | $V_2$ | – | – | 31.2 | 23.4 | – | 35.8 |
| *closed*: Microsoft-Decision-1 | $V_0$ | 35.1 | 29.8 | 33.1 | 25.1 | 36.7 | 25.9 |
| | $V_2$ | 36.1 | 32.4 | 34.6 | 25.5 | 34.3 | 47.2 |

Table 7: Every decider after two rounds of reflection (ten training sentences). UMR: 25 test sentences (five chunks, §4), with the official juːmætʃ and AnCast++'s sentence-graph F (Sun and Xue, 2024); systems tested on all chunks are re-scored on the same five. PARSEME: MWE-based F on the full test sets. Sonnet 5.5 decides through the Claude subscription on UMR and through OpenRouter on PARSEME. English AnCast is averaged over four of the five chunks (it cannot read one gold chunk).

tion improve every full UMR chain (§5), though not on some small two-round changes in English, and they disagree on the comparison with the frontier model: AnCast puts GPT-6.1-sol first in both languages (43.3 and 55.9), ahead of Sonnet (41.1, 51.6) and well ahead of the closed procedures (30–36), while juːmætʃ puts it among the weakest seeds in Chinese. Over all 86 UMR test results, AnCast is 15.8–20.9 points above juːmætʃ for the frontier model, 4–7 above for fragment procedures and 2–3 below for closed ones (`results/ancast_vs_jumatch.json`). The procedures get their alignments from code and the frontier model writes its own, so our advantage over it on UMR is an advantage on the official metric, which rewards well-aligned graphs, and not on graph content alone.

### 5.4 What reflection does

Table 8 summarizes the ten rounds. The reflector lists 8–11 problems per UMR batch and 2–4 per PARSEME batch, and proposes about one edit per problem. In closed mode, measurement keeps about half of the edits (54 of 109 for Jev on Chinese UMR; 13–16 of 36–46 on PARSEME); a typical rejected edit changes nothing measurable ("money is quantity": no change on the batch or the earlier batches), lowers the earlier batches, or is invalid (an edit that broke a YAML file is caught by the loader and never measured). In fragment mode every edit that applies is kept, so the procedure grows faster (Chinese UMR: 36.8k → 50.8k characters despite

| Task | Decisions | problems | edits | kept | compr. | size $V_0$ | size $V_{10}$ |
|---|---|---|---|---|---|---|---|
| UMR zh | Haiku 5.5 | 108 | 57 | 55 | 28 | 36.8k | 50.8k |
| | Jev 1.13 | 108 | 109 | 54 | 0 | 70.5k | 73.7k |
| | Microsoft-Decision-1 | 114 | 106 | 54 | 0 | 70.5k | 74.8k |
| UMR en | Haiku 5.5 | 100 | 59 | 58 | 17 | 36.8k | 51.5k |
| | Jev 1.13 | 87 | 89 | 35 | 0 | 70.5k | 72.6k |
| | Microsoft-Decision-1 | 85 | 71 | 35 | 0 | 70.5k | 72.7k |
| PARSEME zh | GPT-6-luna | 26 | 20 | 18 | 12 | 4.5k | 8.0k |
| | Haiku 5.5 | 29 | 18 | 18 | 9 | 4.5k | 7.5k |
| | Jev 1.13 | 37 | 38 | 16 | 2 | 8.0k | 9.6k |
| | Microsoft-Decision-1 | 38 | 46 | 13 | 0 | 8.0k | 9.2k |
| PARSEME en | GPT-6-luna | 24 | 18 | 18 | 8 | 4.7k | 8.1k |
| | Haiku 5.5 | 26 | 21 | 21 | 9 | 4.7k | 8.1k |
| | Jev 1.13 | 33 | 36 | 13 | 0 | 8.1k | 9.7k |
| | Microsoft-Decision-1 | 38 | 31 | 8 | 0 | 8.1k | 9.2k |

Table 8: Ten rounds of reflection: problems the reflector listed, edits tried (with closed-mode retries), edits kept, compressions, and the size of the editable procedure (characters; closed mode includes code).

28 compressions) than in closed mode (70.5k → 73.7k, mostly code), where edits are short and replace text more often than they add it.

The kept edits read like an annotation manual's errata, and they are where the procedure's improvements come from. In Chinese UMR, Jev's kept edits concern discourse connectives ("limit `and-but`", "speech content is not discourse"), the abstract predicates of non-verbal clauses ("`exist-91` for X中有Y", "`have-place-91` only without a verb"), aspect ("add inceptive aspect") and the person/number of nouns; every one is a readable change to a named step, which an annotator can check and keep or revert. On the training batches themselves, the closed UMR procedures improve steadily: Jev's score on all earlier batches rises from 31.6 after round 1 to 38.9 after round 10, of which 3.7 points carry over to the unseen test documents.

## 6 Discussion

**Diagnosability.** A guideline in context fails silently: GPT-6.1-sol's Chinese UMR graphs are worse than our seeds', and nothing in its output says which of the 366k characters it misread. Each error of a procedure has an address, a step, its input, its options and their probabilities, and that changes how errors are found and fixed. The Haiku seed's English concepts for Chinese words, Jev's unlemmatized English verbs and the closed PARSEME seeds' flood of spurious light-verb constructions (§5.1) were each found from one column of the judge's output and fixed by one edit to one file: by the reflector for the first and the third, and the second is a one-line change to the code that lists the options. The same address makes reflection possible at all: the reflector reads the decisions behind each difference, not just the difference, and its edits are diffs that an annotator can review, keep or revert. We see this as the main practical argument for the approach early in a project, when people and models correct each other.

**Decision models are cheap and promising.** Decision models never write text: they return probabilities over options that code has listed. This makes them the cheapest deciders we tried. Annotating the 100 Chinese UMR test sentences costs $0.28 with Jev and $0.19 with Microsoft-Decision-1, against $1.88 with Haiku in fragment mode and $7.09 for GPT-6.1-sol with the guideline; the 100 PARSEME sentences cost about one cent with either service. A closed parse is fast enough that every edit of reflection can be measured, which is what keeps closed-mode reflection from drifting. And the closed procedure is, as of now, the more accurate one on Chinese UMR (Jev 37.6) and close on English (31.8 against 32.6). Its ceiling is set by the code around it: a decision model cannot pick an option that is not offered (the lemmatizer, §5.1; PARSEME expressions no rule proposes) and cannot write structure. Better enumerators, which are ordinary code, raise that ceiling for every decision model at once, and the transfer result (§5.2) suggests that a procedure learned with one decision model can be handed to another.

**What reflection can and cannot do.** With five gold sentences a round, reflection writes the conventions a guideline leaves implicit (Chinese concepts stay Chinese; 有 is not a light verb; 但 is `but-91`) and fixes procedural errors (a missing candidate kind, a YAML convention). It does so from very little data: the largest gains come in the first rounds (Haiku Chinese UMR +6.9 in round 1; Jev Chinese PARSEME +20.1 in rounds 3–4). Three limits are visible. (1) *Unmeasured edits drift.* In fragment mode we accept every edit that applies, and Haiku's Chinese PARSEME procedure lost almost ten points between rounds 3 and 10 although each edit was locally sensible; measured acceptance gives the closed chains curves that rarely go down. Measuring fragment edits costs a reparse per edit, which a cheap fragment decider would make affordable. (2) *The training batches are tiny.* Closed edits are accepted on changes of a few points on five sentences, and the gap between training and test grows over the rounds (Jev Chinese PARSEME: 88.2 on training batches, 63.4 on test); threshold calibration on the same 50

sentences overfits (§5.2). More gold makes both checks stronger, and the procedure keeps improving as gold arrives. (3) *Reflection edits what exists.* It adds rules, options and questions, but in ten rounds it rarely changed the decomposition itself (which step decides what); in English PARSEME, where the frontier model with the guideline reaches 75.6, the procedures' gains flatten well below it.

**When to decompose.** The frontier model in context is strongest where the scheme is close to what such models already know and the output is a short list, English verbal MWEs; decomposition is strongest where the output is structure and the conventions are particular, Chinese UMR. For a new scheme or language early in a project, the procedure costs a few dollars to run, improves from the first batches of gold, and leaves a trace in which every error can be found.

## 7 Conclusion

We annotated two complex schemes step by step along their guidelines. Compiled into a procedure of rule nodes and small decisions, a guideline that is too long to follow in context becomes something small and cheap models can follow: seeds written from the guideline alone, with no example, match or beat a frontier model reading the whole guideline on three of four task–language pairs, 8–11 points ahead of it on Chinese UMR on the official metric (on graph content alone, measured by AnCast++, the frontier model stays ahead). Every error has an address, so it can be traced to the step that made it, and a reflector can fix the step: from fifty gold sentences, five at a time, the procedures improved in twelve of fourteen runs, by up to 31 points, and on English PARSEME every reflected procedure is above the best shared-task system on the same sentences. Decision models that only return probabilities, at a few cents per hundred sentences, take the best UMR result. The procedure keeps a readable history of what it learned, and the more gold arrives, the more it has to learn from. We think this makes decision orchestration a natural first step for a new annotation project: start from the guideline, annotate with cheap models, and let the procedure improve as the first gold comes in.

## Limitations

**Scale.** Test sets are 95–100 sentences per task and language and training is 50 sentences; each chain was run once, with one document order. With 57–59 gold expressions per PARSEME test set, one expression is worth about one point of F, so differences of a few points there should not be overread; we did not repeat chains with other document orders to measure the variance of reflection. **Unfinished runs.** OpenRouter limits each OpenAI model to 20 requests a minute for our account, and our API budget was fixed. GPT-6-luna as a UMR fragment decider needs about 3,000 calls per test pass, and the OpenAI Decisions chains measure every edit; we therefore ran GPT-6-luna and OpenAI Decisions on UMR, and Sonnet 5.5 as a fragment decider, for two rounds only, tested on 25 sentences (§5.3), and finished fewer rounds for the OpenAI chains on PARSEME (red dashes in the tables). **Data editions.** PARSEME 2.0 has no Chinese data and only an English trial set, so we use editions 1.2 (Chinese) and 1.1 (English); the English 1.1 systems predate pretrained models. There is no UMR 3.0 release; we use the 2026 shared-task data, which is based on UMR 2.1, and the shared task's best result is a placeholder until it is re-scored on our test chunks. **Seeds.** The seeds were written by the authors (with an LLM's help) from the guidelines; their quality affects every result and we have not measured the effort they take. Closed and fragment seeds share the decomposition, so the comparison between modes is not a comparison of decompositions. **Reflection.** Fragment-mode edits are accepted unmeasured and drift; the same reflector (Claude Sonnet 5.5) was used throughout. **Models.** All models are commercial services whose behavior may change; we release the predictions, the procedures at every version, and the cached answers that their terms permit.

## Ethics Statement

We use the UMR shared-task data and PARSEME 1.1 and 1.2 under their licenses (PARSEME 1.2 Chinese: CC BY-NC-SA 4.0) and collect no data from people. The method aims to make the first annotations of a new scheme cheaper and easier to check; it could reduce demand for annotation work, and we see it rather as a way to direct annotators to the decisions that need them. Automatic annotations should be checked before they are used as gold. The runs of this version cost about $59 in logged API charges (decision services $13, LLM calls $46), including runs that were stopped; reflection used a subscription. Generative AI assisted

the implementation, the running of the experiments and the editing of this paper; the authors are responsible for all content.

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## A The seed procedures

All seeds were written from the guidelines before any training sentence was read, and contain no example: no annotated sentence, no example word in an option description, and every fragment skill's `reference/example.md` is empty. They are in `experiment/orchestration-v2/seeds/`.

**UMR.** Both modes share the registry: the sentence wave (`sent_type`, `discourse`, `predicate`, `np_phrase`, `special_entity`, and the rule nodes `coverage` and `antecedent`), the rule wave `reentrancy`, and the `document` wave. Each decision skill restates one part of the guidelines: `sent_type` the criteria for events (Part 3-1-1: processes in any packaging, states and entities only when predicated) and the kinds of pieces; `discourse` the lattice of discourse relations (3-1-6); `predicate` light verbs (3-1-3-5), the abstract predicates of non-verbal clauses and their roles (Tables 4 and 8), word senses (3-1-4), participant and non-participant roles (3-2), the aspect decision procedure (3-3-1: event nominals, then habitual, state, activity, endeavor, performance), polarity (3-3-3) and modal strength (4-3-1); `np_phrase` inverse roles for relative clauses, participant nominalizations and relational nouns (3-2-1-3), modifiers (3-2-2-2), quantity and unit (2-2-5) and person and number (3-3-5); `special_entity` the named-entity types of Table 5, dates and quantities; `document` the temporal superstructure and relations (4-2), the modal dependency (4-3) and coreference (4-1). In fragment mode each is a `skill.md` of 2.5–6.5k characters (35k in all) with the output schema; in closed mode each is a Python file that builds the questions and rules, with the model-facing text in a `prompts.yaml`. Function-word, pronoun and numeral tables are closed-class grammar, not annotation decisions.

**PARSEME.** A rule node proposes candidates from the input's dependency tree: in English a verb with its particle (`compound:prt`), with the head noun of its object or oblique, or with the adposition of its oblique; in Chinese a multi-character verb token, a verb with the verb or adjective right after it that depends on it, a verb with its object, and four-character tokens. In closed mode, each candidate is asked the yes/no tests of its kind from the guidelines' decision trees (LVC.0–LVC.5, VID.1–VID.5, VPC.1–VPC.3, IAV, and the Chinese MVC tree), and rules walk the tree on the answers with one threshold. In fragment mode the LLM reads a `decide.md` that restates the categories and tests, sees the candidates, and returns the expressions of the sentence, which may include expressions the rules did not propose.

## B Reflection

**Algorithm 1** Reflection on five gold sentences per round.

**Require:** seed $V_0$, training batches $B_1, \ldots, B_{10}$, mode $\in$ {closed, fragment}
**for** $r = 1, \ldots, 10$ **do**
  $V \leftarrow V_{r-1}$; parse $B_r$ with $V$; judge; attribute each difference to a step
  $P, E \leftarrow$ REFLECTOR(gold, prediction, differences, decisions, files of $V$, history)
  **for** each edit $e \in E$ **do**
    **if** $e$ applies, the procedure loads, and $e$ copies no test text **then**
      **if** mode = closed **then**
        $\Delta_b, \Delta_g \leftarrow$ change of F on $B_r$ and on $B_1..B_{r-1}$ with $e$
        keep $e$ iff $(\Delta_b > 0 \wedge \Delta_g \geq -0.5) \vee (\Delta_g > 0 \wedge \Delta_b \geq 0)$
      **else** keep $e$
      **end if**
    **end if**
  **end for**
  closed: send rejected edits back once with their measured effect; try the rewrites
  **for** each file $f$ longer than $1.5\times$ its seed **do**
    rewrite $f$ shorter (reflector); closed: keep it iff no set falls by more than 0.5
  **end for**
  $V_r \leftarrow V$ (frozen; tested on the test set)
**end for**

The reflector's instructions (abridged): the procedure follows a written guideline, code cuts, enumerates and assembles, decision nodes judge; list every problem you can find, however small, with its cause in the procedure; propose edits (`replace`, `insert_after`, `delete`, each an exact anchor in one file) that fix them on these sentences and on unseen text; prefer general rules and better option descriptions to special cases; keep every file valid; when you add a rule, delete or merge what it makes redundant. The closed-mode variant lists the useful kinds of edits (the right answer among the options with a rule branch; sharper option descriptions; questions added, split or removed; thresholds; conventions written into rules; code that cuts or assembles wrongly), the fragment-mode variant its own (conventions, misleading instructions removed, ex-

plicit decision procedures, output format). The full prompts are in `code/reflect.py`.

## C Data details

**UMR.** Splits are frozen by `code/splits.py` (seed 20261012) in `experiment/orchestration-v2/data/`. Chinese training: one batch of five consecutive sentences from each of ten training documents among `chinese_umr-0001`–`0018`; Chinese test: 20 chunks of five consecutive sentences from the 13 blind-test documents with gold (`0021`–`0024`, `0027`–`0035`), at least one chunk per document. English training: five, three and two batches from `english_umr-0004`, `0578` and `0579`; English test: all 95 sentences of `english_umr-0005`–`0008` in 20 chunks. Input tokens are the gold `Words` lines, as in the shared task. Gold is rewritten in the strict format the official scorer reads (one graph per sentence, alignments restricted to graph variables, document-level relations restricted to the chunk). Scoring: `compare_umr.pl -tsv`, with and without `-no-document-level`.

**PARSEME.** Chinese: PARSEME 1.2 `ZH`; English: PARSEME 1.1 `EN`. Training batches: three sentences with at least one VMWE and two without, ten batches. Test: 50 sentences with and 50 without a VMWE, drawn from the official test file. Unseen F uses a training file reduced to the sentences with a VMWE, which gives the same seen/unseen split as the full file and runs in seconds. The shared-task systems' released predictions are re-scored on the same 100 sentences with the official `evaluate.py`.

**Guidelines.** The UMR guidelines (`umr4nlp/umr-guidelines`, `guidelines.md`, 366k characters) and the PARSEME 1.2 guidelines (335k characters) are given to the baseline in full and were the only source of the seeds.

## D All test scores

## E Costs

## F Earlier experiments: fragment decisions by Claude Sonnet 5.5

Before the experiments of the main text, we ran the fragment-mode UMR procedure with a stronger parser, Claude Sonnet 5.5, on Chinese UMR, with an earlier hand-written seed that contained worked examples, an earlier test protocol (whole documents; AnCast++ instead of the shared task's metric) and unmeasured reflection by Claude Opus 5.5 on one whole training document per round. These numbers are not comparable with the main text (different seed, test documents, metric and gold exposure) and are reported for completeness.

One training document raised the first test set from 45.3 to 50.8 F and a second to 52.6, with temporal relations improving most; third and fourth documents left the test scores flat. Unmeasured reflection accepted 440 of 441 edits and the skills grew from 41k to 156k characters in ten rounds, which motivated the length compression of the main text.

| Task | Decisions | $V_0$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | $V_6$ | $V_7$ | $V_8$ | $V_9$ | $V_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UMR zh | Haiku 5.5 | 20.9 | 27.8 | 30.5 | 31.1 | 31.9 | 32.4 | 31.1 | 33.5 | 33.6 | 33.5 | 33.2 |
| | Jev 1.13 | 33.9 | 35.6 | 35.9 | 36.8 | 36.7 | 36.5 | 36.3 | 37.3 | 37.2 | 37.2 | 37.6 |
| | OpenAI Decisions | 32.1 | | | | | | | | | | |
| | Microsoft-Decision-1 | 35.0 | 34.8 | 35.1 | 35.1 | 34.8 | 35.4 | 35.2 | 35.0 | 34.9 | 34.9 | 35.8 |
| | Jev + cascade | 35.5 | | | | | | | | | | 38.2 |
| UMR en | Haiku 5.5 | 28.8 | 29.4 | 29.5 | 30.5 | 31.4 | 31.1 | 31.3 | 30.8 | 30.7 | 30.6 | 32.6 |
| | Jev 1.13 | 29.8 | 30.1 | 31.0 | 30.8 | 31.0 | 31.3 | 31.7 | 31.4 | 31.0 | 31.6 | 31.8 |
| | OpenAI Decisions | 26.2 | | | | | | | | | | |
| | Microsoft-Decision-1 | 29.0 | 29.1 | 30.4 | 30.6 | 30.1 | 31.1 | 31.3 | 30.8 | 30.2 | 31.0 | 31.0 |
| | Jev + cascade | 31.4 | | | | | | | | | | 32.2 |
| PARSEME zh | GPT-6-luna | 62.5 | 62.5 | 67.8 | 67.2 | 72.9 | 73.4 | 74.1 | 70.6 | 69.2 | 70.6 | 72.5 |
| | Haiku 5.5 | 59.5 | 57.1 | 62.9 | 62.9 | 53.1 | 51.6 | 47.2 | 54.7 | 47.2 | 51.6 | 53.2 |
| | Jev 1.13 | 37.8 | 36.0 | 36.0 | 52.2 | 56.1 | 56.1 | 56.1 | 56.1 | 56.1 | 56.1 | 63.4 |
| | Microsoft-Decision-1 | 36.7 | 36.7 | 34.3 | 39.4 | 40.0 | 40.0 | 46.0 | 46.0 | 46.0 | 46.0 | 46.0 |
| | Jev + cascade | 39.8 | | | | | | | | | | 53.4 |
| PARSEME en | GPT-6-luna | 62.3 | 61.2 | 57.6 | 60.7 | 55.5 | 60.8 | 58.1 | 59.5 | 55.3 | 59.8 | 58.4 |
| | Haiku 5.5 | 57.6 | 57.4 | 55.9 | 59.5 | 58.8 | 61.3 | 61.1 | 60.0 | 56.6 | 57.9 | 58.2 |
| | Jev 1.13 | 22.5 | 36.5 | 38.4 | 39.7 | 40.0 | 45.9 | 53.2 | 53.7 | 53.7 | 53.7 | 53.7 |
| | OpenAI Decisions | 38.0 | 33.9 | 35.8 | | 41.9 | | | | | | |
| | Microsoft-Decision-1 | 25.9 | 47.2 | 47.2 | 47.2 | 47.2 | 47.2 | 46.7 | 46.7 | 46.7 | 44.0 | 43.5 |
| | Jev + cascade | 25.8 | | | | | | | | | | 44.1 |

Table 9: Test F1 of every frozen version (UMR: juːmætʃ; PARSEME: MWE-based). Blank: not tested (budget).

| | UMR | | PARSEME | |
|---|---|---|---|---|
| Decisions | zh | en | zh | en |
| GPT-6.1-sol + guideline | 7.09 | 5.09 | 4.05 | 4.02 |
| Haiku 5.5 (fragment) | 1.88 | 0.83 | 0.03 | 0.02 |
| GPT-6-luna (fragment) | – | – | 0.01 | 0.01 |
| Jev 1.13 (closed) | 0.28 | 0.10 | 0.01 | 0.01 |
| Microsoft-Decision-1 (closed) | 0.19 | 0.07 | 0.01 | 0.01 |

Table 10: API cost (USD) of annotating the test set (95–100 sentences) with the seed procedure, uncached, as billed through OpenRouter. Reflection used Claude Sonnet 5.5 through a subscription; its list-price equivalent is $0.4–3.2 per ten-round chain.

| System | Refl. docs (sent.) | T1: 0004–05 F/ Comp. | T2: 0005–07 F/ Comp. |
|---|---|---|---|
| Single call, Sonnet 5.5 | 0 | 32.7 / 27.7 | 29.4 / 24.9 |
| $R_0$ (seed), Sonnet 5.5 | 0 | 45.3 / 44.0 | – |
| $R_{13}$ | 1 (51) | 50.8 / 49.4 | – |
| $R_{23}$ | 2 (111) | 52.6 / 51.2 | – |
| $R_{32}$ | 3 (142) | 52.6 / 51.1 | 55.2 / 52.1 |
| $R_{42}$ | 4 (201) | n/a | 55.6 / 52.2 |

Table 11: Earlier Chinese UMR runs with Sonnet 5.5 fragment decisions (AnCast++ sentence-graph F and comprehensive score, macro-averaged over documents). $R_{xy}$: the version after round $y$ on training document $x$. "n/a": 0004 was a training document by then.

| Doc | seed | iter 1 | iter 2 | iter 3 | after |
|---|---|---|---|---|---|
| 0001 | 40.3 | 40.3 | 59.6 | 66.9 | 70.9 |
| 0002 | 50.6 | 58.9 | 66.3 | 70.4 | 73.4 |
| 0003 | 49.7 | 57.5 | 69.5 | – | – |
| 0004 | 39.6 | 49.7 | 57.9 | – | – |

Table 12: The same runs: sentence F on each training document over the rounds of reflection on it. Training documents gain far more than test documents, as in the main text.